Do microbial communities separate by habitat? A re-analysis of the Global Patterns 16S dataset

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1 Summary

The Global Patterns dataset (Caporaso et al., 2011) sequenced the 16S rRNA gene from environmental and host-associated samples, and the original paper reported that communities split between host-associated and free-living habitats. Using 23 non-control samples (18,773 OTUs; three mock-community controls excluded), I test that split directly. Habitat (host-associated vs free-living) explains 12% of Bray-Curtis dissimilarity and 22% of weighted-UniFrac distance (PERMANOVA; $p = 0.001$ for both, the smallest value 999 permutations can give), while the finer sample type explains 69% and 77%. The result is the same without rarefying (12%, $p = 0.001$). At phylum level, DESeq2 (Love et al., 2014) finds Firmicutes, Tenericutes and Fusobacteria strongly enriched in host-associated samples. With 2 to 4 samples per sample type, the tests are underpowered for any single type, and the dispersion assumption behind PERMANOVA is borderline (see "Is the PERMANOVA result trustworthy?").

2 Question and data

Do the microbial communities of different habitats group by habitat, and which taxa distinguish host-associated from free-living communities? The data are the `GlobalPatterns` object distributed with the Bioconductor package `phyloseq` (McMurdie & Holmes, 2013): an OTU table with 19,216 OTUs and 26 samples, a taxonomy table and a phylogenetic tree. Sample types and counts after removing the mock communities: Feces (4), Freshwater (2), Freshwater (creek) (3), Ocean (3), Sediment (estuary) (3), Skin (3), Soil (3), Tongue (2). Sequencing depth ranges from 58,688 to 2,357,181 reads per sample.

3 Methods

Rarefaction. For alpha and beta diversity, samples were subsampled without replacement to the smallest library (58,688 reads), which leaves 12,386 OTUs. Rarefying throws away data and is criticised for differential abundance testing (McMurdie & Holmes, 2014), so I use it only for diversity, run a non-rarefied sensitivity check, and never for the DESeq2 analysis.

Alpha diversity. Observed OTUs and Shannon index. Sample types compared with a Kruskal-Wallis test; habitats with a Wilcoxon rank-sum test.

Beta diversity. Bray-Curtis dissimilarity and weighted UniFrac distance, ordinated by principal coordinates analysis (PCoA). Group differences tested by PERMANOVA (`vegan::adonis2`, 999 permutations) with habitat or sample type as the single predictor. Homogeneity of dispersion was checked with `betadisper`, because PERMANOVA can return a significant result from unequal spread alone.

Differential abundance. Counts were aggregated to phylum, phyla with fewer than 1,000 total reads dropped, and DESeq2 fitted with `~ habitat` (free-living as reference) using `poscounts` size factors, which suit sparse count tables. Adjusted $p < 0.05$ (Benjamini-Hochberg).

Software. R 4.5.3, phyloseq 1.54.2, vegan 2.7.3, DESeq2 1.50.2.

4 Results

4.1 Sequencing depth and alpha diversity

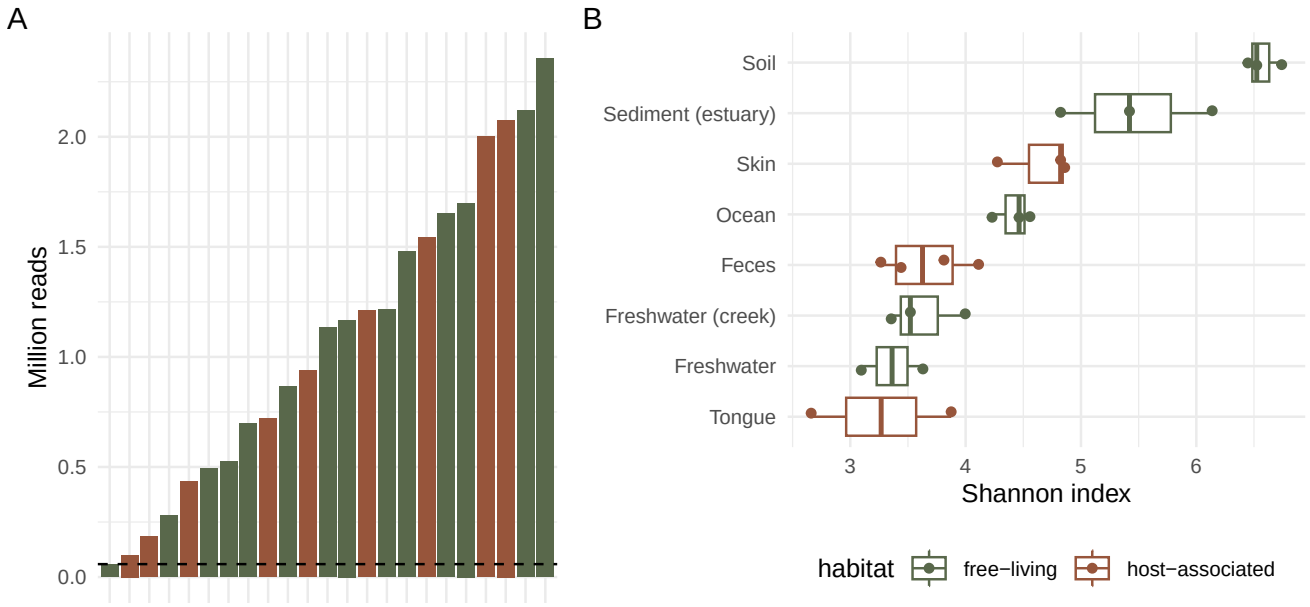


Figure 1: (A) Reads per sample before rarefaction (dashed line = rarefaction depth, barely visible at this scale). (B) Shannon diversity after rarefaction, by sample type.

Library sizes vary about 40-fold, from 58,688 to 2,357,181 reads (Figure 1A), which is why diversity is compared at a common depth. Mean Shannon diversity is 4.78 for free-living and 3.9 for host-associated samples (Wilcoxon $p = 0.16$, not significant at this sample size). Shannon diversity differs among the 8 sample types in a Kruskal-Wallis test ($p = 0.0076$), but with 2 to 4 samples per type, pairwise comparisons would be unreliable and I do not report them.

4.2 Community composition

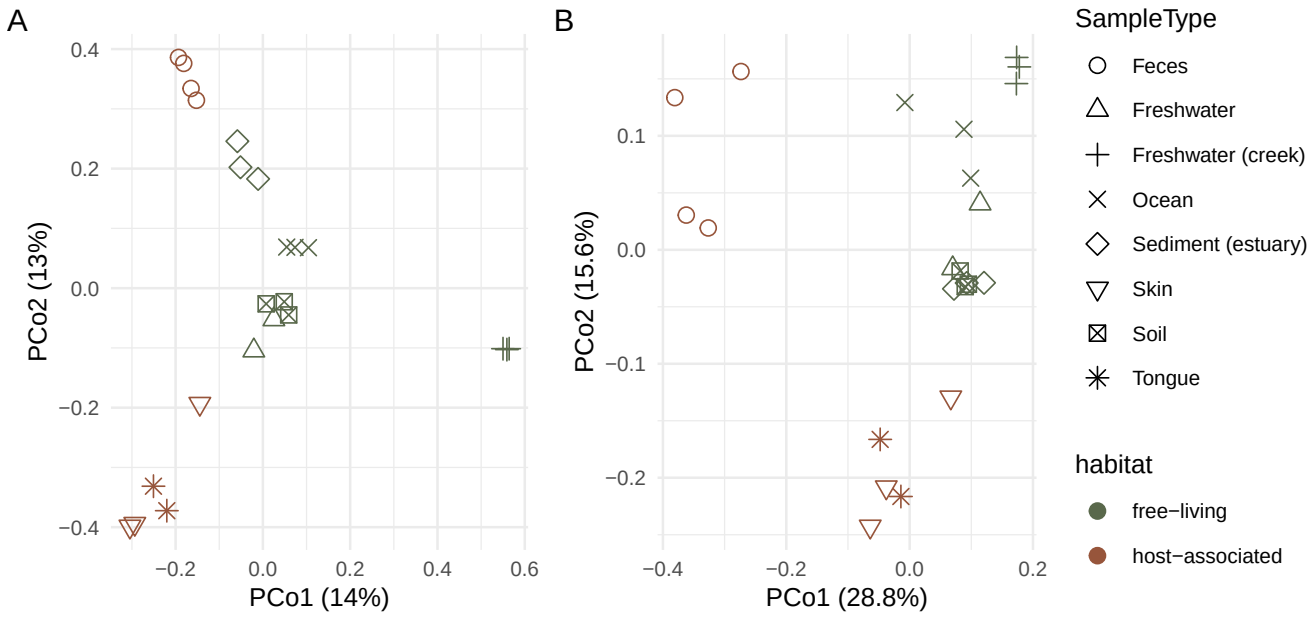


Figure 2: PCoA of (A) Bray-Curtis dissimilarity and (B) weighted UniFrac distance on rarefied data. Colour = habitat, shape = sample type.

Table 1: PERMANOVA (999 permutations). R^2 is the share of total dissimilarity explained by the predictor.

Distance	Predictor	R^2	Pseudo-F	p
Bray-Curtis	Habitat (2 groups)	0.119	2.84	0.001
Bray-Curtis	Sample type (8 groups)	0.690	4.78	0.001
Weighted UniFrac	Habitat (2 groups)	0.216	5.80	0.001
Weighted UniFrac	Sample type (8 groups)	0.769	7.14	0.001

Samples of the same type mostly sit together in both ordinations (Figure 2; one skin sample sits apart from the other two), and sample type explains most of the variation (Table 1). Habitat alone explains less, because the free-living group contains very different environments (soil, ocean, freshwater, sediment). Weighted UniFrac, which accounts for phylogenetic relatedness and abundance, gives a larger habitat effect than Bray-Curtis (22% against 12%). That is consistent with differences in deeper lineages as well as in which OTUs are present, but I did not test that directly.

4.3 Is the PERMANOVA result trustworthy?

PERMANOVA assumes similar within-group spread. Dispersion differs between habitats at $p = 0.065$ (`betadisper`, Bray-Curtis), which is borderline: part of the Bray-Curtis habitat difference could reflect unequal spread rather than a shift in centroid. Two things limit the worry. The weighted-UniFrac result, which is less sensitive to rare OTUs, is stronger, and the non-rarefied relative-abundance version gives $R^2 = 0.12$ ($p = 0.001$). Neither removes the caveat; it is a reason to describe the effect as a difference in composition and dispersion together.

4.4 Which phyla distinguish host-associated communities?

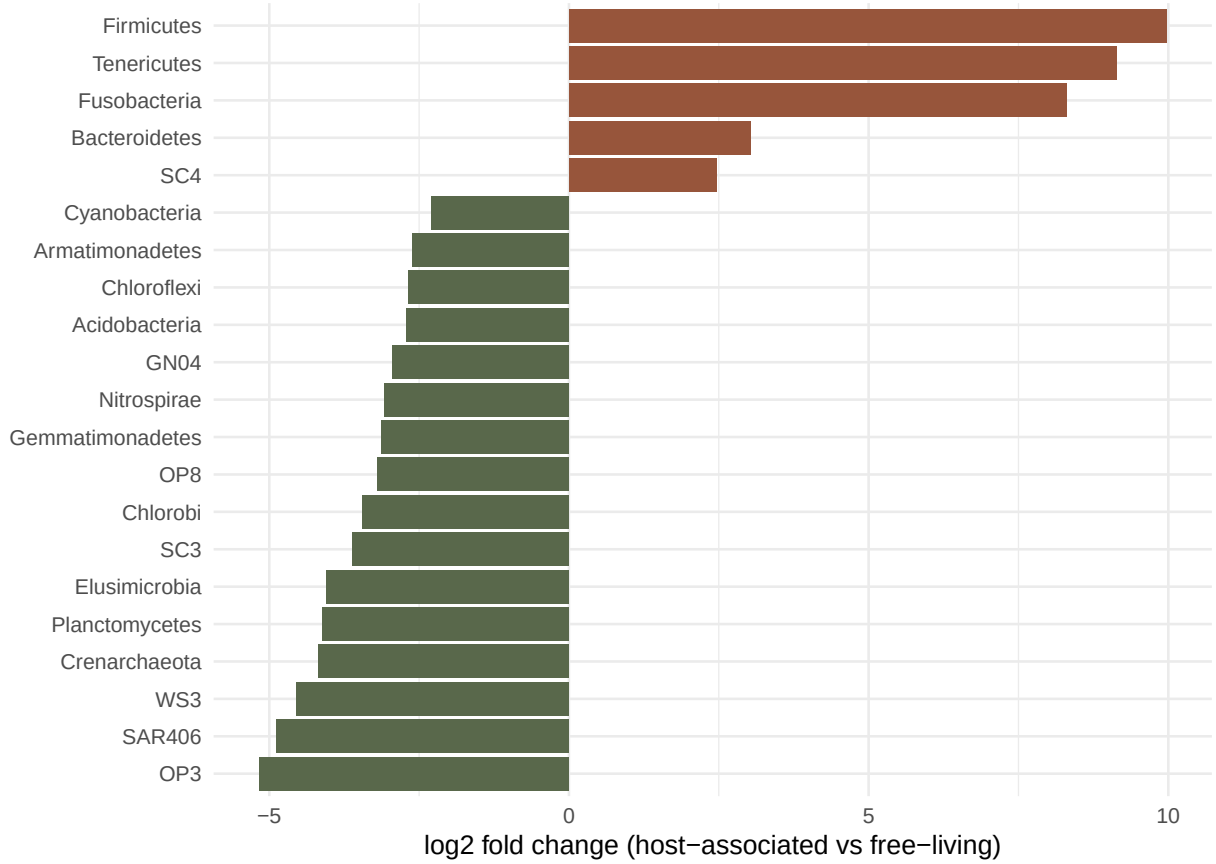


Figure 3: Phyla with adjusted p < 0.05: log2 fold change in host-associated relative to free-living samples (DESeq2, poscounts size factors).

Of 34 phyla tested, 21 differ at adjusted $p < 0.05$: 5 are higher in host-associated samples and 16 higher in free-living samples (Figure 3). The strongest host-associated enrichments are Firmicutes, Tenericutes and Fusobacteria. The host-associated group is only gut, skin and tongue (9 samples). Log2 fold changes as large as the ones seen here (up to about 10) usually arise when a phylum is nearly absent in one group; I did not check that for each phylum, so I read the ranking as qualitative and the fold changes as imprecise.

5 Discussion and limitations

Replication is very small. There are 2 to 4 samples per sample type and 9 versus 14 per habitat. Group-level tests are valid in principle but have little power, so non-significant results here are not evidence of no difference.

The groups are not independent replicates. The habitats combine different environments, and samples from the same site type may share a source. The test treats each sample as independent evidence for "host-associated vs free-living", which overstates how much independent information the data contain.

Rarefaction. Used only for diversity, with a non-rarefied check. Dropping to 58,688 reads per sample discards most of the data for deeply sequenced samples.

Phylum level. Aggregating to phylum makes the count table well-behaved for DESeq2 but hides the OTU- and genus-level changes that usually matter in practice.

Scope. This reproduces a qualitative finding of the original paper on a dataset already prepared in R; it does not involve the raw reads, OTU picking or taxonomy assignment.

6 Provenance and reproducibility

The data are public and shipped with Bioconductor `phyloseq` (from Caporaso et al. (2011)). The analysis code is this document (`report/report.qmd`). It was written and run with the assistance of Claude Code (Anthropic); I am responsible for checking and understanding every result before relying on it. No part derives from university coursework or assessed material.

References

Caporaso, J. G., Lauber, C. L., Walters, W. A., Berg-Lyons, D., Lozupone, C. A., Turnbaugh, P. J., Fierer, N., & Knight, R. (2011). Global patterns of 16S rRNA diversity at a depth of millions of sequences per sample. *Proceedings of the National Academy of Sciences*, *108*(Supplement 1), 4516–4522. https://doi.org/10.1073/pnas.1000080107

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